CSE443 : Bioinformatics : Term Project Proposal

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Topic: Deep Learning Methods for Single-Cell RNA-seq Clustering and Embedding — A Comparative Review

Problem Statement

Single-cell RNA sequencing (scRNA-seq) lets researchers measure gene expression at the level of individual cells instead of averaging over a whole tissue, which is a big step up for understanding cell heterogeneity. But the data it produces is extremely high-dimensional, sparse (a lot of genes show zero counts just from technical dropout, not because they aren't expressed), and noisy. Clustering cells into meaningful types or states is one of the first and most important steps in any scRNA-seq pipeline, and it affects pretty much every downstream conclusion a researcher draws, from identifying rare cell populations to studying disease mechanisms.

Over the last few years, dozens of deep learning-based clustering and embedding methods have been proposed to deal with these challenges, and they use very different underlying ideas: autoencoders, variational autoencoders, graph neural networks, contrastive learning, attention mechanisms, and more recently transformer-based foundation models like scGPT. What's missing is a clear picture of how these approaches actually compare to each other in terms of assumptions, the kind of data they were tested on, and where each one falls short. This project brings together a set of recent papers in this space and compares them critically instead of treating each one in isolation.

Motivation

I picked this topic mainly because scRNA-seq clustering is a fairly mature, well-benchmarked area of computational biology, which makes it realistic to review carefully in one semester, but it's still moving fast enough that a synthesis is actually useful right now. New papers keep coming out (I found several from just 2025 and early 2026 while doing initial reading) that each claim to beat the previous generation of methods, and it's hard to tell from reading one paper alone whether a new architecture is solving a real limitation or just doing well on the specific benchmark it was tested on.

Cell-type identification is also foundational to a lot of downstream biomedical work, drug target discovery, disease subtyping, developmental biology, so understanding which deep learning approaches are actually reliable, and under what conditions, has practical value beyond just academic interest. Doing a comparative review instead of picking one narrow technical sub-problem lets me engage with a much wider range of ideas in the field, which I think will be more useful for my own understanding going forward.

Proposed Approach

I'll start by reading two or three published review papers on scRNA-seq clustering (e.g. Zhang et al., 2023, RNA; Peng et al., 2020, RNA Biology) to get a feel for how these reviews are usually structured and how deep the comparisons typically go, since the goal is a synthesis and not a paper-by-paper summary.

For the main review, I plan to select around 8-9 recent papers (mostly 2023-2026) that represent different methodological families, graph-based autoencoders, contrastive/siamese clustering, adversarial graph learning, and transformer-based foundation models, so the comparison has some real contrast to it rather than comparing near-identical methods. For each paper I'll note the problem it addresses, the model/architecture used, the datasets and evaluation metrics reported (ARI, NMI, and silhouette score come up constantly), and its stated strengths and limitations. I'll then build a comparison table across these dimensions and use it to pin down where the papers actually disagree or leave gaps, for example in generalization across tissue types, computational scalability, or interpretability of the learned embeddings, and close with a short discussion of what future work in this space could look like.